



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Alexandru Balica
December, 9th, 2025



Outline

1. **Executive Summary**
2. **Business Problem & Objective**
3. **Data Collection & Methodology**
4. **Exploratory & Geospatial Analysis**
5. **Predictive Modeling**
6. **Key Findings & Limitations**
7. **Appendix & Technical Resources**

Executive Summary

Objective

- **Predict whether a Falcon 9 first stage will land successfully and identify operational patterns associated with landing outcomes.**

Approach

- Collected launch data through a REST API and web scraping, then cleaned and engineered features for analysis.
- Used Python, SQL, interactive visualizations, and geospatial analysis to investigate launch-site, payload, orbit, and booster patterns.
- Built and tuned Logistic Regression, Decision Tree, KNN, and SVM classifiers using cross-validation.

Key Findings

- First-stage landing success improved substantially across later missions.
- Launch outcomes varied across orbit types, payload ranges, launch sites, and booster configurations.
- Geospatial analysis showed launch facilities positioned near coastlines and major transportation infrastructure.
- All four classifiers achieved **0.833 test accuracy**, but the small test set and weaker identification of failed landings limit how strongly model performance should be interpreted.

Outcome

- **Delivered an end-to-end analytical workflow combining data acquisition, exploratory analysis, interactive visualization, and predictive modeling.**

Business Problem & Objective

Business Context

- Falcon 9 first-stage recovery is a key component of SpaceX's reusable-launch model.
- Understanding the factors associated with successful landings can support operational analysis and improve estimates of launch economics.

Project Objective

Build an end-to-end data science workflow to:

- identify patterns associated with first-stage landing outcomes;
- analyze relationships among launch site, orbit, payload, booster configuration, and flight history;
- explore launch-site geography and proximity to infrastructure;
- train and compare classification models that predict whether the first stage will land successfully.

Key Questions

- Which mission and booster characteristics are most associated with successful landings?
- How did landing reliability change across flights and booster generations?
- Can landing outcome be predicted reliably from the available mission features?

Section 1

Methodology

Methodology Overview

1. Data Acquisition

- Retrieved Falcon 9 launch records from public APIs and structured datasets.
- Scraped supplementary launch information from web sources.

2. Data Preparation

- Cleaned missing and inconsistent values.
- Standardized fields and engineered analysis-ready features.

3. Exploratory Analysis

- Used Python visualizations and SQL queries to identify patterns across payloads, sites, orbits, boosters, and outcomes.

4. Geospatial & Interactive Analysis

- Built Folium maps to examine launch-site geography and infrastructure proximity.
- Created a Plotly Dash dashboard for interactive exploration.

5. Predictive Modeling

- Trained Logistic Regression, Decision Tree, KNN, and SVM classifiers.
- Tuned hyperparameters with GridSearchCV and evaluated performance using cross-validation and classification metrics.

Data Collection Pipeline

Data Sources

- SpaceX launch data from public APIs, CSV datasets, and web sources.
- Mission-level information included launch date, site, payload, orbit, booster configuration, and landing outcome.

Pipeline

- Acquire raw launch data
- Convert records into structured Pandas DataFrames
- Standardize schemas across sources
- Clean missing and duplicate records
- Engineer analysis-ready features
- Prepare standardized datasets for the project's analytical stages

Outcome

Analysis-ready datasets covering mission, booster, payload, orbit, launch-site, and landing information.

Raw Sources



API / CSV / Web Scraping



Pandas DataFrames



Cleaning & Standardization



Feature Engineering



Analysis-Ready Dataset

Data Collection – SpaceX API

Source

Public SpaceX REST API containing structured launch and rocket metadata.

Process

- Sent HTTP requests using Python requests.
- Parsed JSON responses into Pandas DataFrames.
- Extracted relevant mission, payload, rocket, launch-site, and outcome fields.
- Standardized fields for downstream analysis.

Technical Skills

Python · REST API · JSON · requests · Pandas

SpaceX REST API



HTTP Request



JSON Response



Pandas DataFrame



Selected & Standardized Features

Data Collection - Scraping

Source

Public launch tables containing supplementary Falcon 9 mission information.

Process

- Retrieved HTML pages using Python requests.
- Parsed table structure with BeautifulSoup.
- Extracted mission-level fields from HTML tables.
- Converted records into structured Pandas DataFrames.
- Cleaned and standardized fields for downstream analysis.

Why Use a Second Source?

Web scraping supplemented structured API data and demonstrated an alternative data-acquisition workflow for semi-structured web content.

Technical Skills

Python · BeautifulSoup · HTML Parsing · Pandas

Web Page / HTML Table



requests



BeautifulSoup Parsing



Extracted Records



Pandas DataFrame

Data Wrangling & Feature Preparation

Cleaning

- Standardized column names and data types.
- Handled missing and inconsistent values.
- Removed duplicate or irrelevant records.

Transformation

- Converted raw fields into analysis-ready numeric and categorical variables.
- Harmonized schemas across API, scraped, and provided datasets.

Feature Engineering

- Created the binary first-stage landing target.
- Prepared launch-site, orbit, payload, booster, and flight-history features.
- Encoded categorical variables for machine-learning models.

Output

Clean, encoded features prepared for classification, with standardized datasets supporting the exploratory and visualization stages.

Raw Sources



Schema Standardization



Missing / Duplicate Handling



Feature Engineering



Categorical Encoding



Model-Ready Dataset

Exploratory Data Analysis

— Visualization

Objective

- Identify patterns between first-stage landing outcomes and key mission characteristics.

Analysis Focus

- Landing success rate across orbit types
- Flight number vs. landing outcome
- Payload mass vs. landing outcome
- Launch site differences
- Changes in landing success over time

Why It Matters

- EDA was used to identify candidate relationships, detect temporal trends, and guide feature selection before predictive modeling.

Technical Skills

- Pandas · Matplotlib · Seaborn

Exploratory Data Analysis – SQL

Objective

Use SQL to answer structured analytical questions directly from the launch dataset.

Examples

- Compared launch outcomes across launch sites.
- Aggregated landing outcomes using GROUP BY.
- Filtered missions by payload ranges and launch characteristics.
- Analyzed booster versions associated with successful landings.
- Ranked outcome frequencies across selected time periods.

Why SQL

SQL provided a fast, reproducible way to validate dataset structure and answer targeted business questions independently of the Python analysis.

Technical Skills

SQL · Filtering · Aggregation · GROUP BY · Sorting

Geospatial Analysis — Folium

Objective

Examine how launch sites relate spatially to coastlines and transportation infrastructure.

Map Features

- Launch-site markers with landing outcomes
- Marker clustering for readability
- Distance measurements to nearby infrastructure
- Polygons connecting launch sites to coastlines, highways, railways, and cities

Analytical Value

The map added geographic context by visualizing launch-site proximity to coastlines and selected transportation features.

Technical Skills

Folium · Geospatial Analysis · Interactive Mapping

Interactive Dashboard — Plotly Dash

Objective

Create an interactive interface for exploring landing outcomes without requiring users to inspect code.

Dashboard Features

- Overall landing-success distribution
- Launch-site filtering
- Payload-range filtering
- Dynamic chart updates based on user selections
- Interactive exploration of site and payload relationships

Business Value

The dashboard translated analytical results into a stakeholder-friendly format for rapid exploration and comparison.

Technical Skills

Plotly Dash · Interactive Visualization · Python

Predictive Modeling – Classification

Objective

Predict whether the Falcon 9 first stage will land successfully using mission and launch characteristics.

Models Compared

- Logistic Regression
 - Decision Tree
- K-Nearest Neighbors
- Support Vector Machine

Modeling Workflow

- Prepare numerical and categorical features
- Split data into training and test sets
- Apply preprocessing to training data
- Tune hyperparameters with GridSearchCV
- Evaluate candidate models using cross-validation
- Compare final performance on the held-out test set

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Result

All four classifiers achieved **0.833 test accuracy**, but the small test set limits how strongly differences—or similarities—between models can be interpreted

Results Summary

Exploratory Findings

- First-stage landing success improved across later flights.
- Landing outcomes varied across orbit types and payload ranges.
- Launch sites exhibited different observed landing-success rates.

Geospatial Findings

- Launch facilities are concentrated near coastlines and major transportation infrastructure.
- Spatial analysis quantified distances from launch sites to selected geographic and transportation reference points.

Modeling Findings

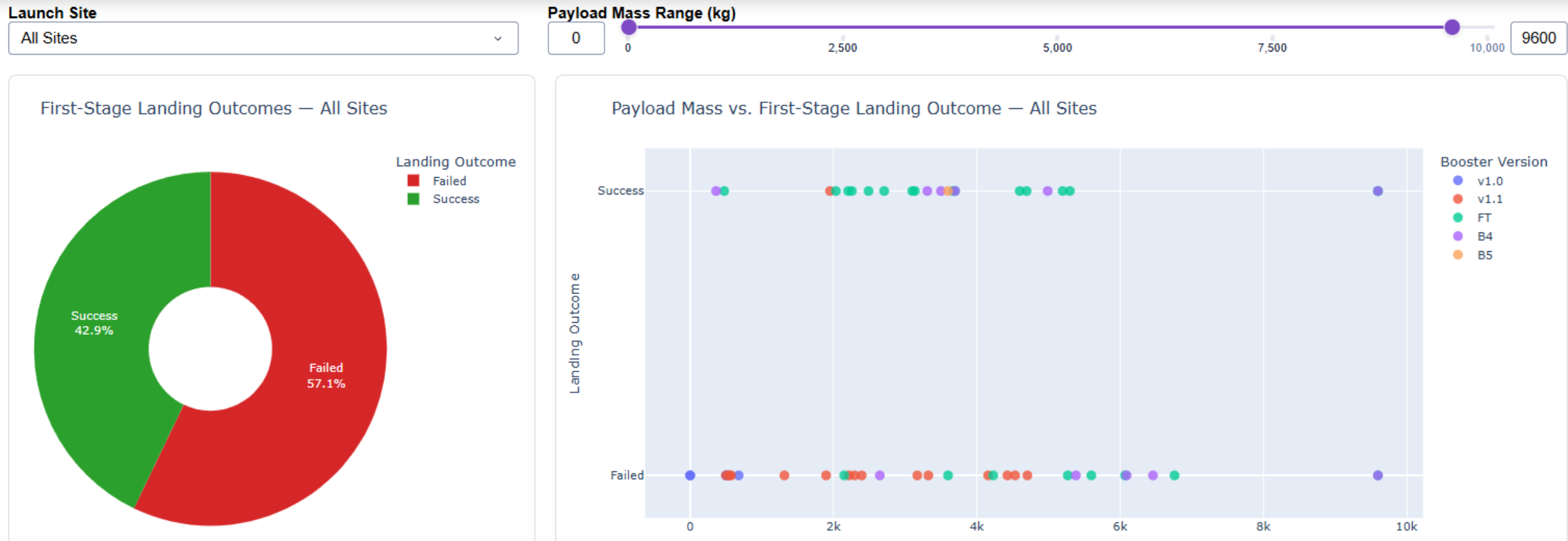
- Logistic Regression, Decision Tree, KNN, and SVM all achieved **0.833 test accuracy**.
- The confusion matrix showed that successful landings were identified more reliably than failed landings.
- The small evaluation sample means model comparisons should be interpreted cautiously.

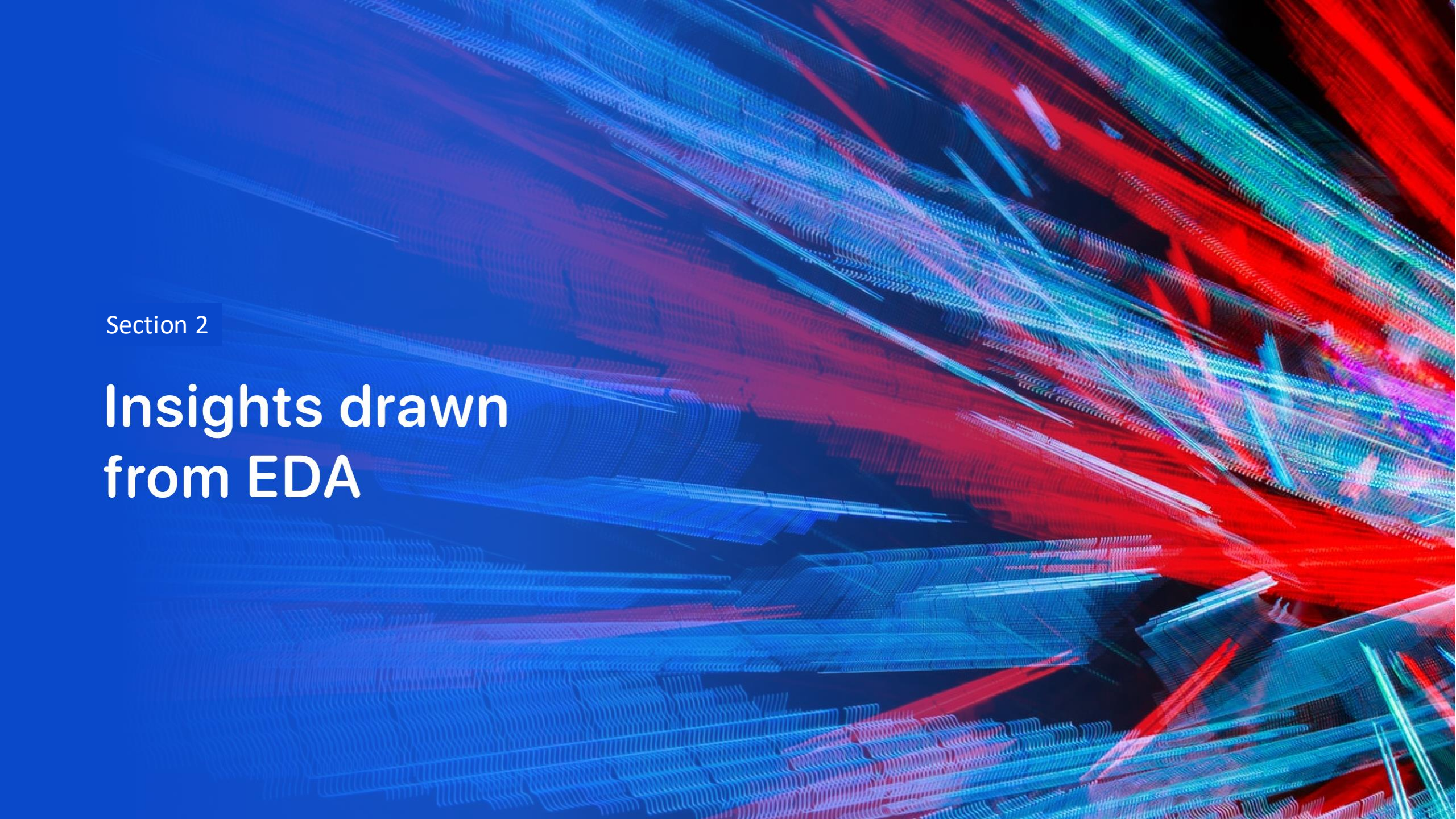
Main Takeaway

The analysis suggests that flight history, booster generation, mission profile, and launch context are associated with first-stage landing outcomes, while the predictive results demonstrate a workable classification pipeline rather than a production-ready model.

Interactive Dashboard — Example View

- **Dashboard capabilities**
- Filter landing outcomes by launch site.
- Explore payload mass ranges interactively.
- Compare landing-success distributions and payload relationships dynamically.

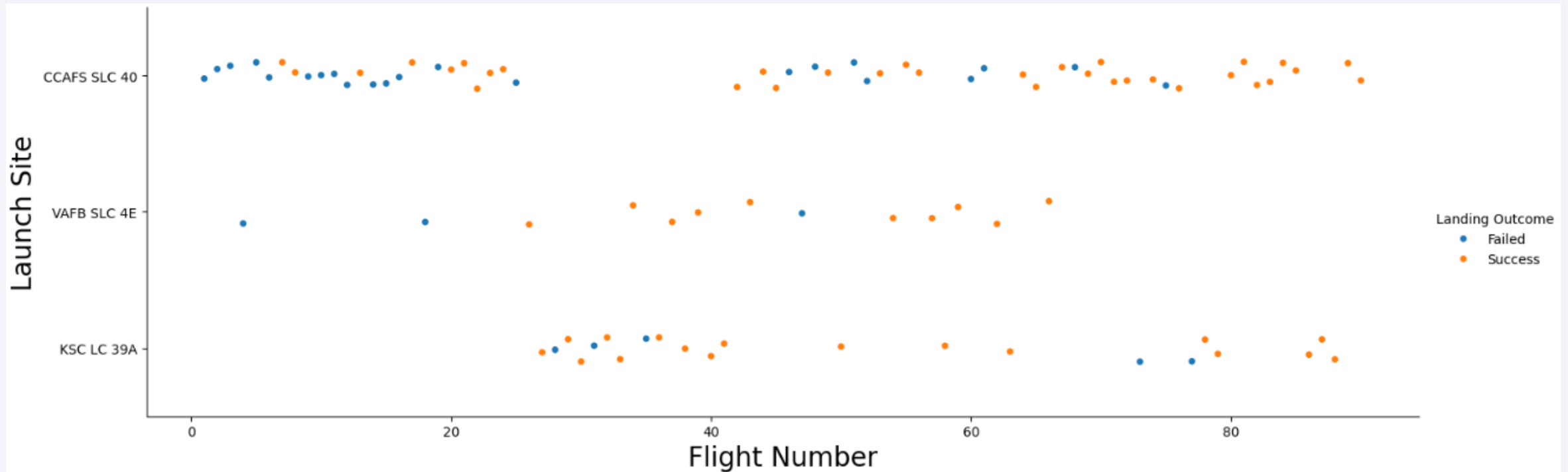


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

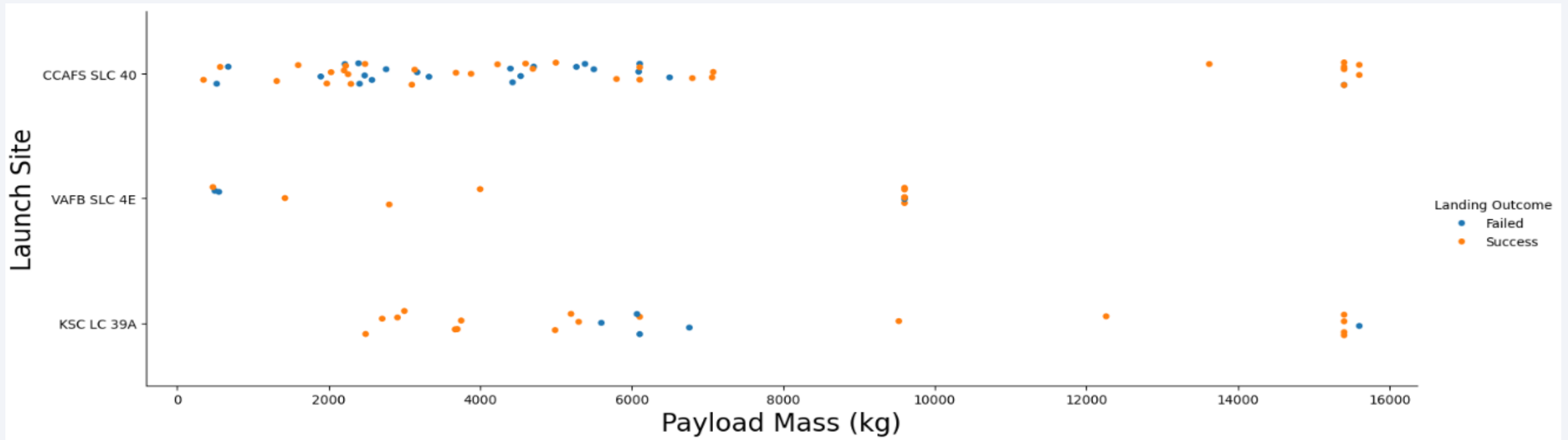
Insights drawn from EDA

Flight Number vs. Launch Site by Landing Outcome



Later flights show a higher concentration of successful first-stage landings across launch sites. Flight number also captures changes in booster generation, mission characteristics, and operational experience over time, so the relationship should be interpreted as an association rather than a direct causal effect.

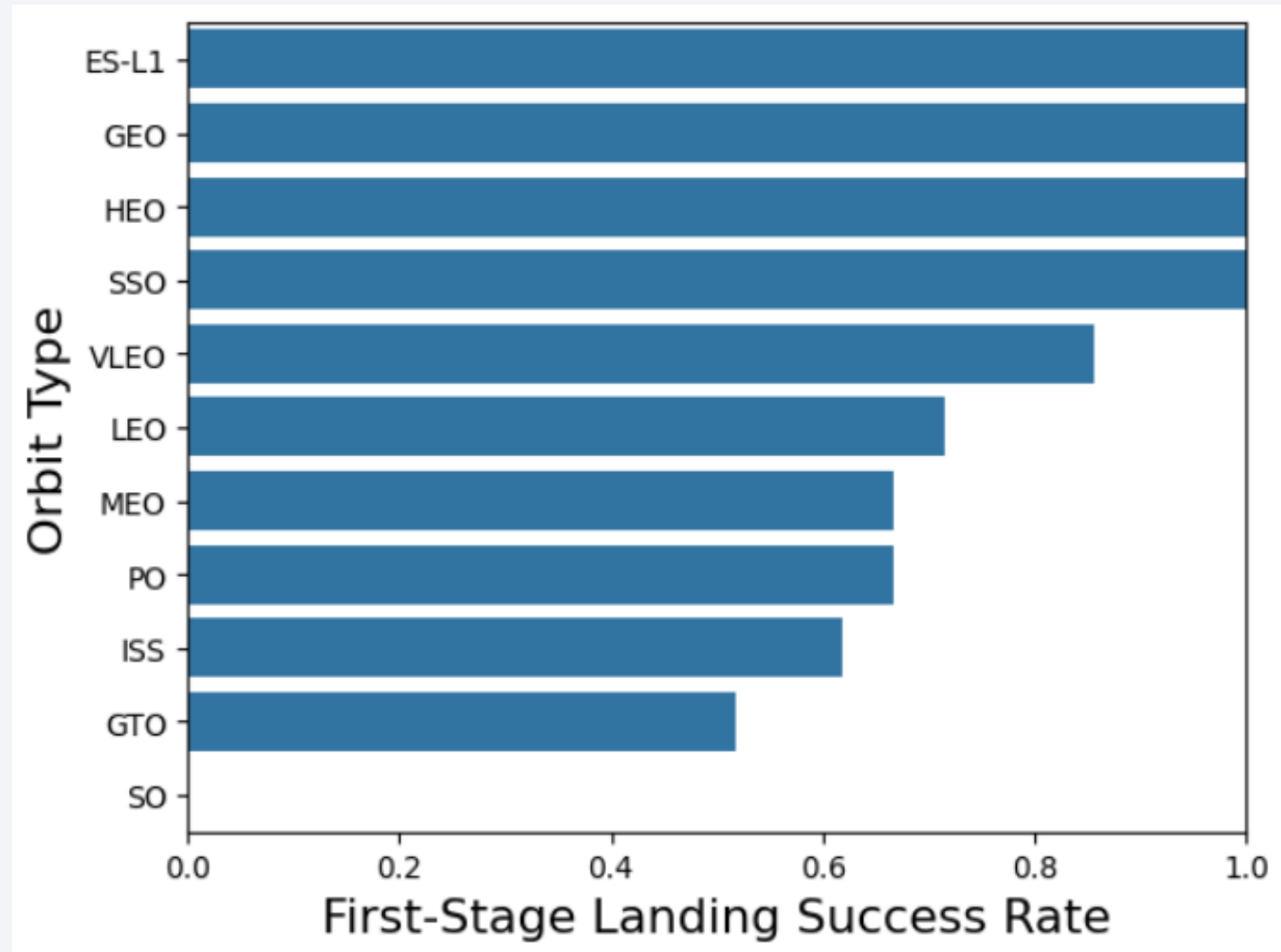
Payload Mass vs. Launch Site by Landing Outcome



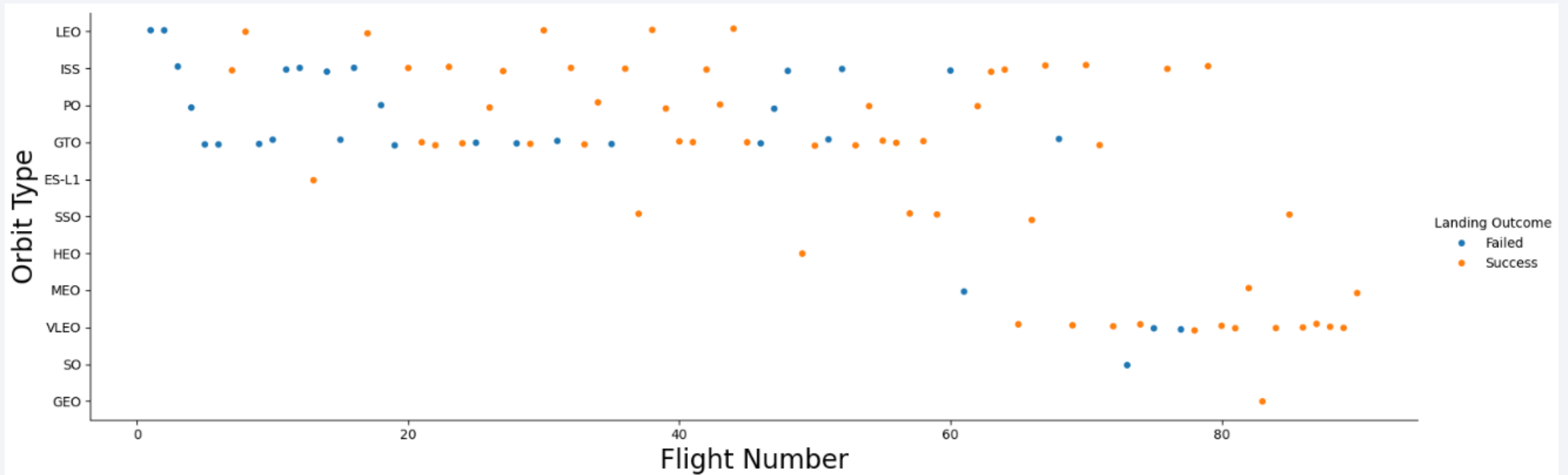
Landing outcomes vary across launch sites and payload ranges. Successful landings occur across a broad range of payload masses, while VAFB SLC-4E contains no missions above roughly 10,000 kg in this dataset. Differences between sites should be interpreted descriptively, since payload, booster generation, orbit, and launch period are also related.

First-Stage Landing Success Rate by Orbit Type

Landing success rates vary substantially across orbit types in this dataset. ES-L1, GEO, and HEO show the highest observed rates, while GTO is notably lower. These results are descriptive and should be interpreted cautiously because some orbit categories contain relatively few missions.

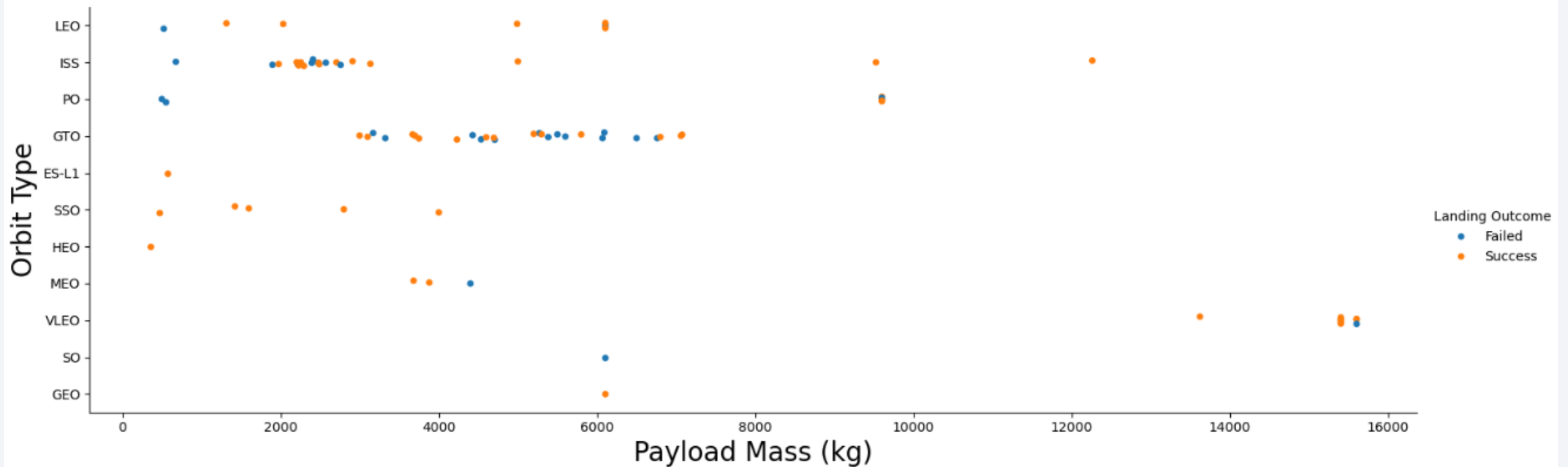


Flight Number vs. Orbit Type by Landing Outcome



Successful landings become more common in later flights across several orbit categories. However, the pattern varies by orbit, and flight number also captures broader changes over time such as booster generation and mission mix.

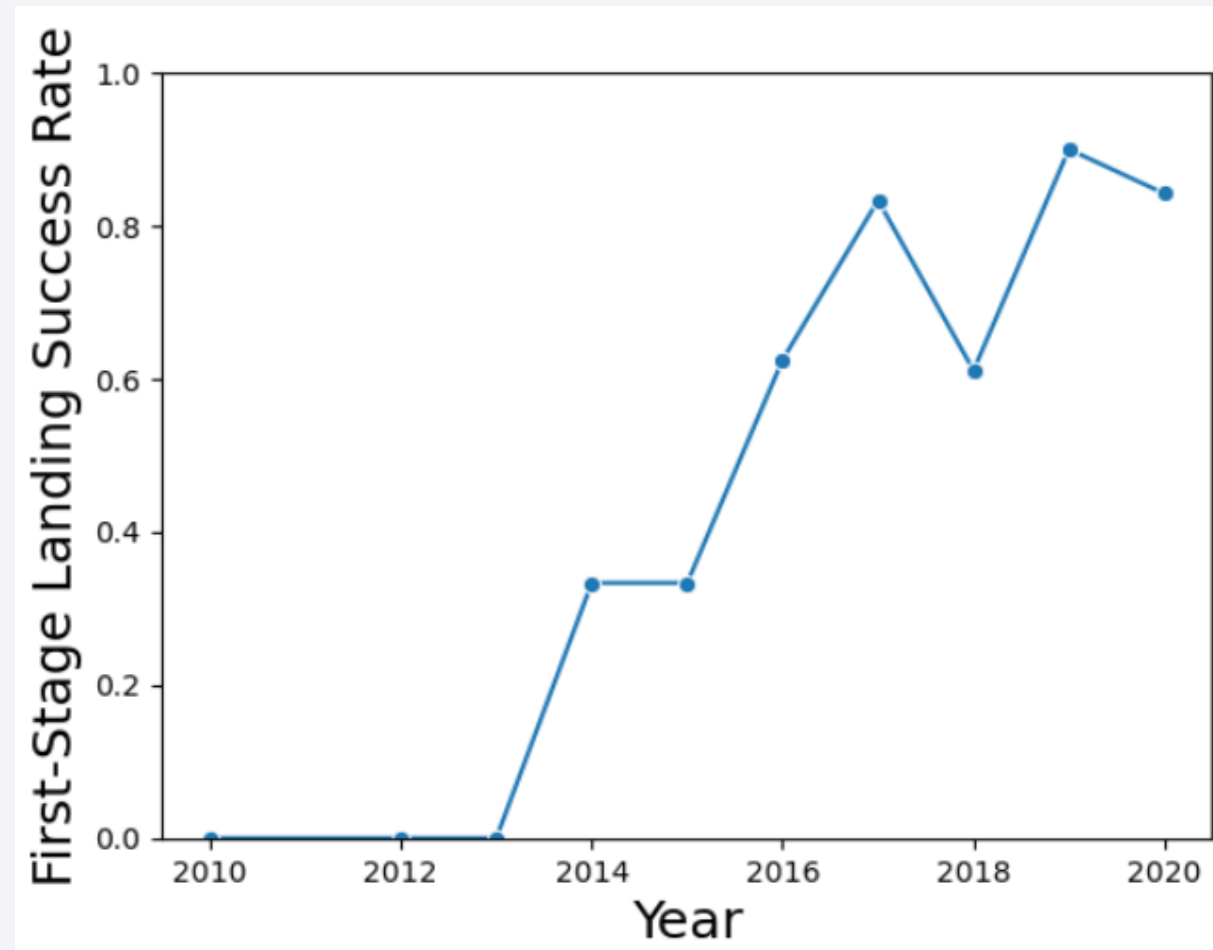
Payload Mass vs. Orbit Type by Landing Outcome



Landing outcomes vary across both payload mass and orbit type. Successful landings occur across a wide payload range, while some orbit categories show mixed outcomes at similar payload levels. Because orbit, payload, booster generation, and mission period are related, these patterns should be interpreted as associations rather than isolated effects.

First-Stage Landing Success Rate by Year

First-stage landing success improved markedly over time, rising from no successful landings in the earliest years to consistently high rates in later years. The year-to-year variation reflects a relatively small number of missions in some periods, so the trend is more informative than any single annual value.



Launch Sites Identified with SQL

```
SELECT DISTINCT Launch_Site  
FROM SPACEXTABLE;
```

Results

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

Explanation:

- Using `DISTINCT`, the query identifies four launch-site labels represented in the dataset. These sites form the basis for later comparisons of landing outcomes and mission characteristics.

Filtering Launch Sites with SQL

```
SELECT Launch_Site  
FROM SPACEXTABLE  
WHERE Launch_Site LIKE 'CCA%'  
LIMIT 5;
```

Results

- CCAFS LC-40
- CCAFS LC-40
- CCAFS LC-40
- CCAFS LC-40
- CCAFS LC-40

Explanation

The `LIKE 'CCA%'` filter selects launch records associated with Cape Canaveral site labels, while `LIMIT 5` restricts the displayed output to five records.

Total Payload Mass for CRS Missions

```
SELECT SUM(PAYLOAD_MASS__KG_) AS TotalPayloadMass  
FROM SPACEXTABLE  
WHERE Payload LIKE '%CRS%';
```

Result

- $\approx 111,268$ kg

Explanation

- The query sums payload mass for missions whose payload description contains “CRS,” returning a total of 111,268 kg in this dataset.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS MeanPayloadMass  
FROM SPACEXTABLE  
WHERE Booster_Version LIKE '%F9 v1.1%';
```

Result

- $\approx 2,535$ kg

Explanation

- F9 v1.1 missions in this dataset carried an average payload mass of approximately 2,535 kg.

First Successful Ground-Pad Landing

```
SELECT MIN(Date)
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Success (ground pad)';
```

Result

Date: 2015-12-22

Explanation

Using MIN(Date) on records with Landing_Outcome = 'Success (ground pad)', the earliest successful ground-pad landing in this dataset occurred on December 22, 2015.

Successful Drone-Ship Landings with 4,000–6,000 kg Payloads

```
SELECT Booster_Version  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ > 4000  
AND PAYLOAD_MASS__KG_ < 6000  
AND Landing_Outcome = 'Success (drone ship)';
```

Results

- F9 FT B1026
- F9 FT B1022
- F9 FT B1021.2
- F9 FT B1031.2

Explanation

The query identifies booster versions that successfully landed on a drone ship while carrying payloads between 4,000 and 6,000 kg.

Mission Outcome Distribution

```
SELECT
    TRIM(Mission_Outcome) AS Mission_Outcome,
    COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
GROUP BY TRIM(Mission_Outcome)
ORDER BY Outcome_Count DESC;
```

Results

• Mission_Outcome	Outcome_Count
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Explanation

- Grouping by `Mission_Outcome` provides the actual count of each mission outcome category rather than only the total number of records.

Booster Versions Carrying the Maximum Payload

Results

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
SELECT MAX(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE);
```

Explanation

- A subquery first identifies the maximum payload mass in the dataset, then returns every booster version associated with that maximum value. Multiple results indicate that several missions share the same maximum payload mass.

Failed Drone-Ship Landing Attempts in 2015

```
SELECT
    substr(Date, 6, 2) AS Month,
    Booster_Version,
    Launch_Site,
    Landing_Outcome
FROM SPACEXTABLE
WHERE substr(Date, 1, 4) = '2015'
    AND Landing_Outcome = 'Failure (drone ship)';
```

Results

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Explanation

Two failed drone-ship landing attempts appear in the 2015 records: one in January and one in April, both using Falcon 9 v1.1 boosters from CCAFS LC-40.

Landing Outcome Distribution: 2010–2017

Landing_Outcome	OutcomeCount	SELECT Landing_Outcome, COUNT(*) AS OutcomeCount FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY OutcomeCount DESC;
No attempt	10	
Success (drone ship)	5	
Failure (drone ship)	5	
Success (ground pad)	3	
Controlled (ocean)	3	
Uncontrolled (ocean)	2	
Failure (parachute)	2	
Precluded (drone ship)	1	

Explanation

- “No attempt” was the most common outcome in this period (10 records). Successful and failed drone-ship landings were tied at five each, illustrating the transition from early missions without recovery attempts toward active booster-recovery operations.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

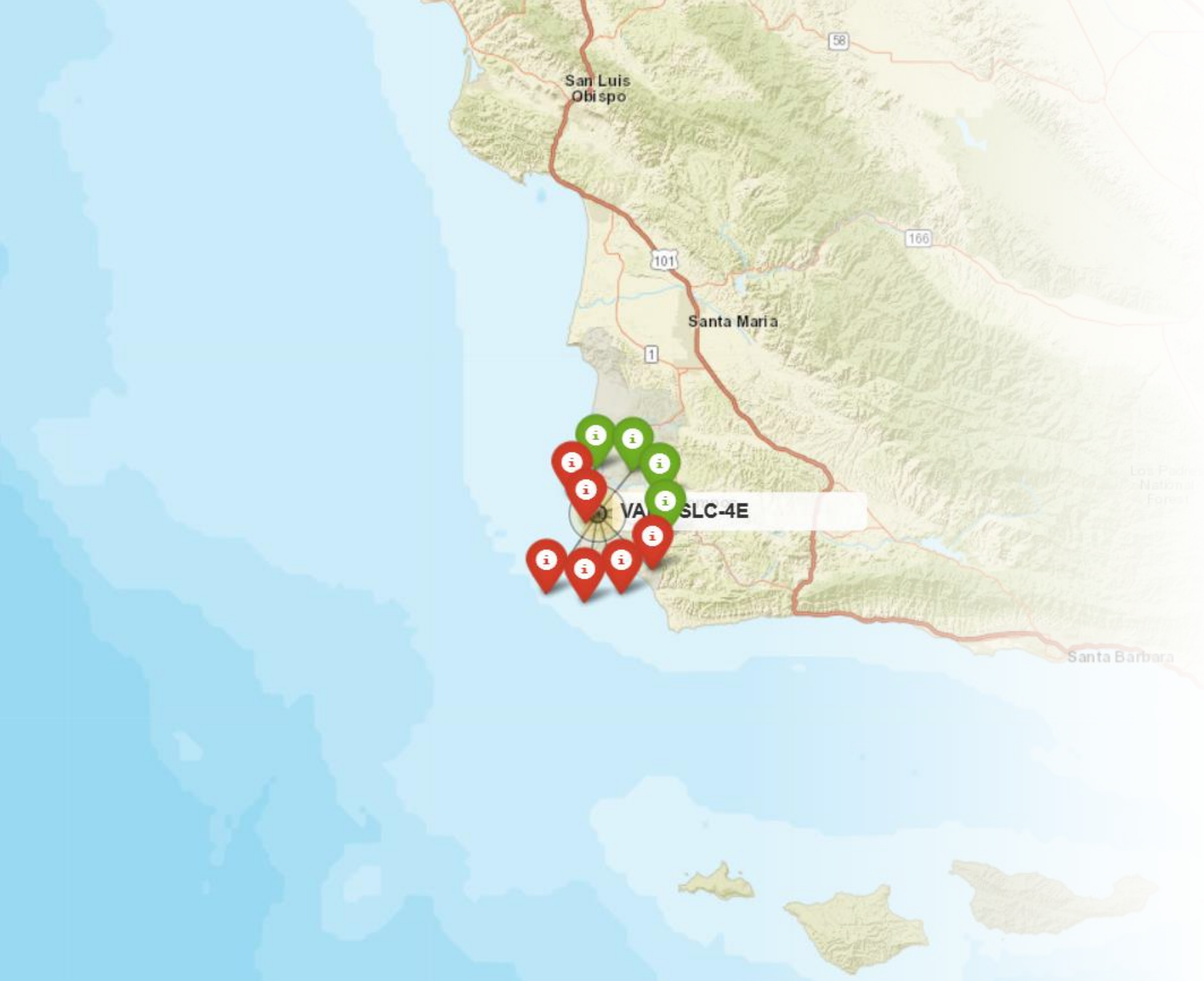


Geographic Distribution of SpaceX Launch Sites

- Launch-site records in the dataset are concentrated in two U.S. regions: Florida's Space Coast and Vandenberg in California. This geographic view provides context for the subsequent proximity analysis of coastlines and transportation infrastructure.

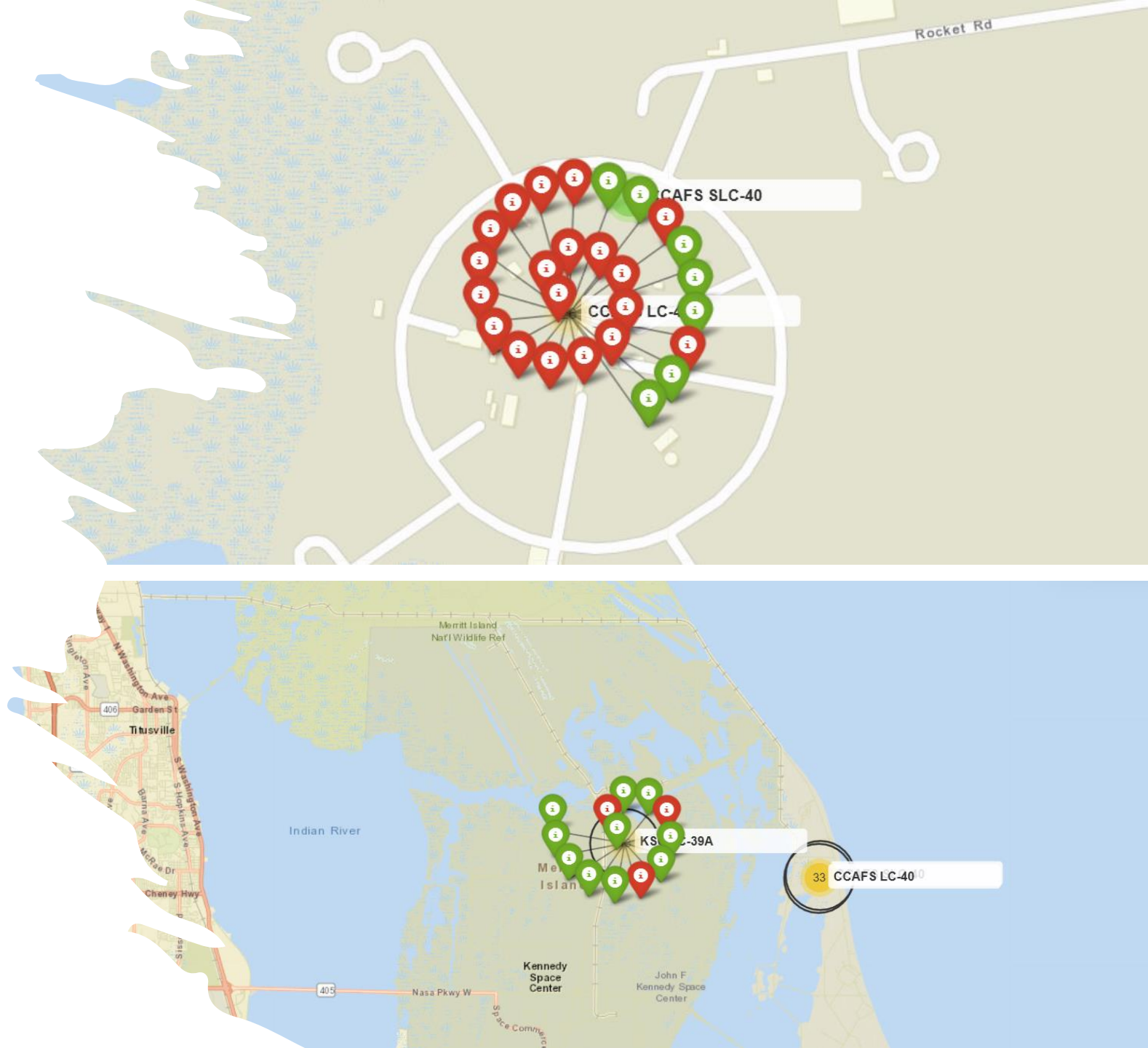
First-Stage Landing Outcomes – Vandenberg

Individual launch records at VAFB SLC-4E are color-coded by first-stage landing outcome: green for successful landings and red for failures. Clustering allows overlapping launch records at the same site coordinates to be explored interactively.



First-Stage Landing Outcomes — Florida Space Coast

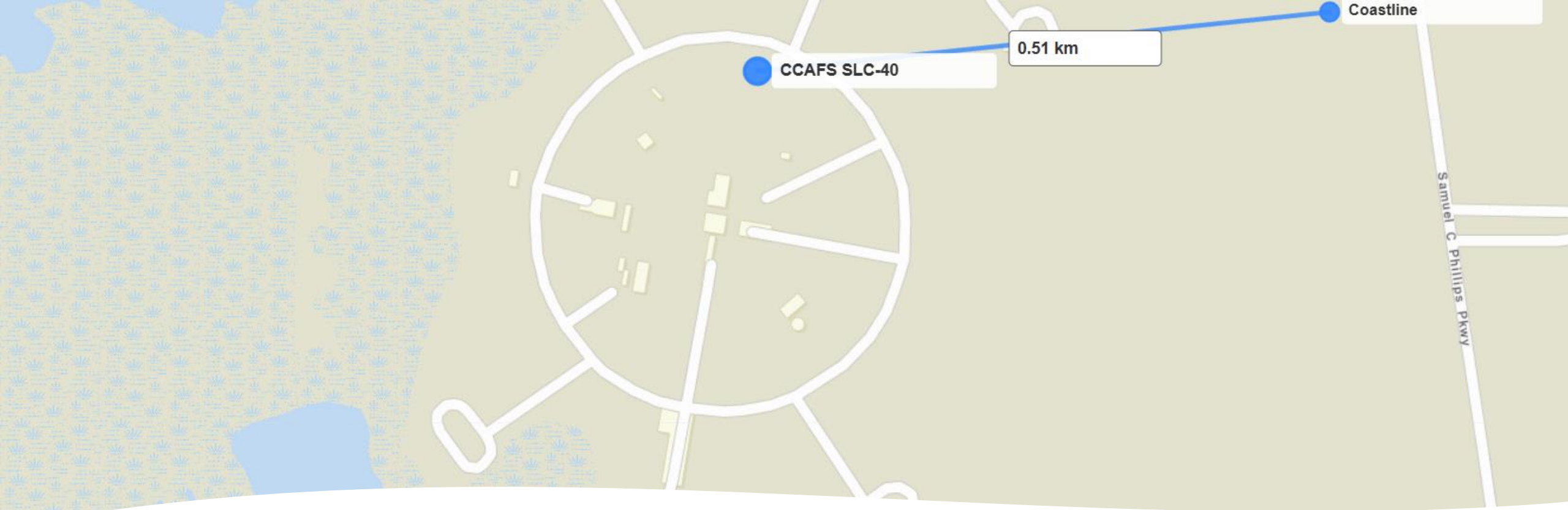
Launch records from Cape Canaveral and Kennedy Space Center are color-coded by first-stage landing outcome. Green markers represent successful landings and red markers represent failures, while clustering helps separate overlapping records at closely located launch sites.



Landing Outcome Patterns Across Launch Sites

- Key Elements & Findings to Highlight
- **Color coding:** Green markers represent successful first-stage landings; red markers represent failures.
- **Marker clustering:** Multiple launch records share identical site coordinates, so clustering keeps the map readable.
- **Geographic pattern:** Launch activity is concentrated at Florida and California sites.
- **Outcome pattern:** The mix of successful and failed landings varies across launch sites and can be compared interactively.
- **Interpretation:** The map provides a spatial comparison of landing outcomes, but it does not by itself establish why one site performs differently from another.

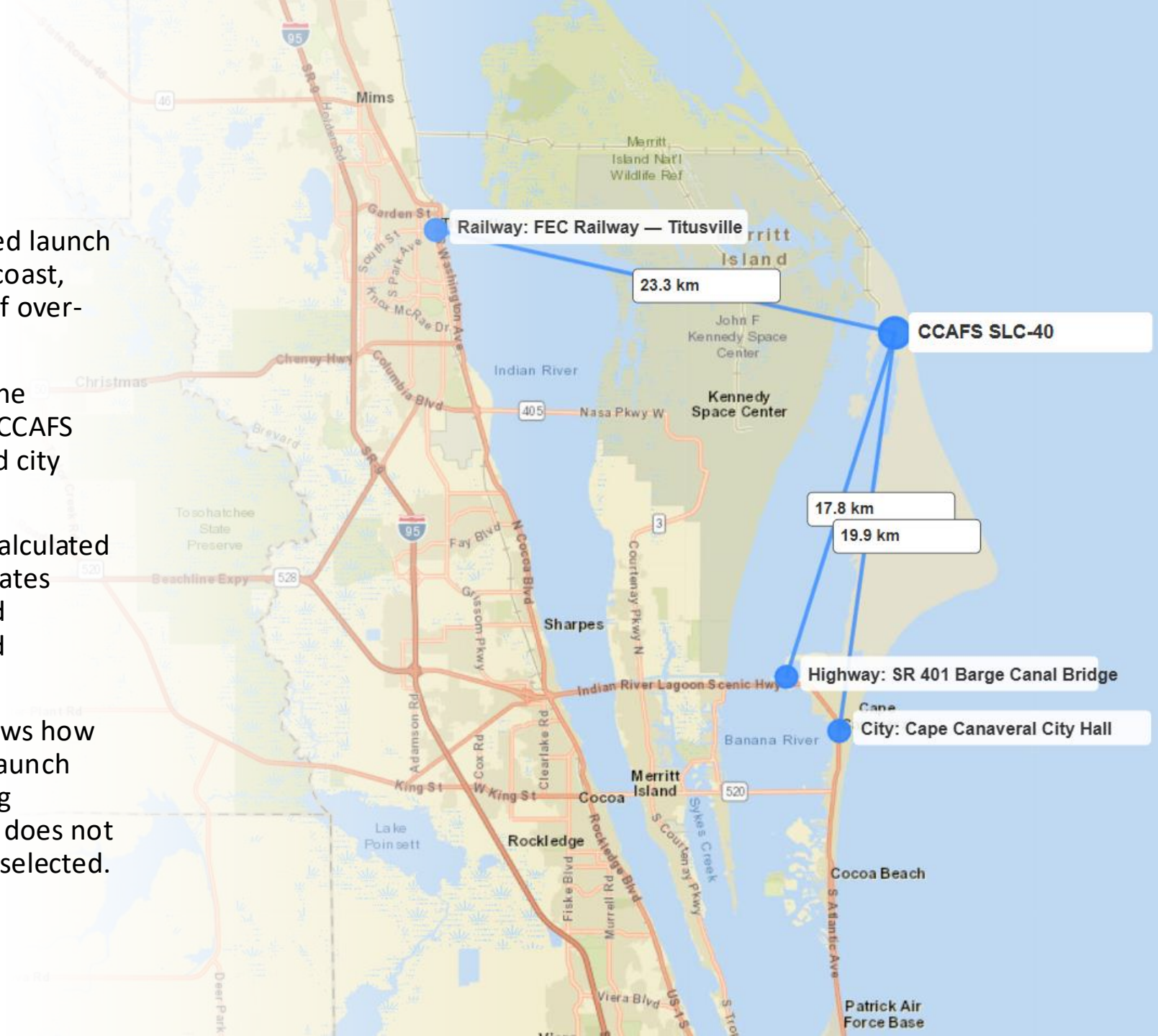


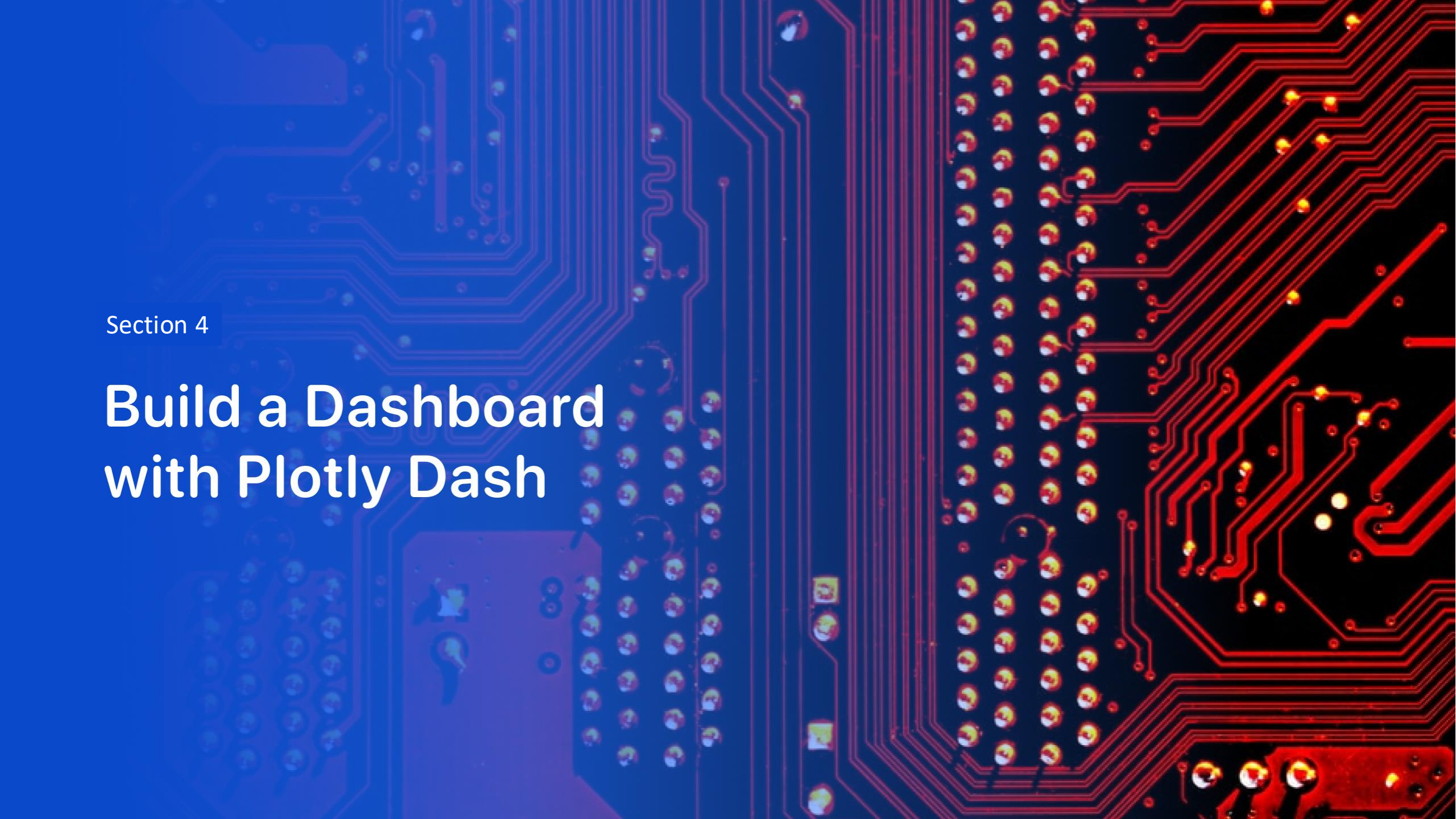


Launch Site Proximity Analysis

Launch Site Proximity — Key Findings

- **Coastline proximity:** The selected launch site is located very close to the coast, supporting direct visualization of over-water launch corridors.
- **Representative POIs:** Straight-line distances were calculated from CCAFS SLC-40 to selected rail, road, and city reference points.
- **Method used:** Distances were calculated from latitude/longitude coordinates using the Haversine formula and displayed with map markers and polylines.
- **Interpretation:** The analysis shows how geospatial tools can quantify a launch site's relationship to surrounding infrastructure and geography; it does not by itself prove why the site was selected.





Section 4

Build a Dashboard with Plotly Dash

KSC LC-39A

0

0

2,500

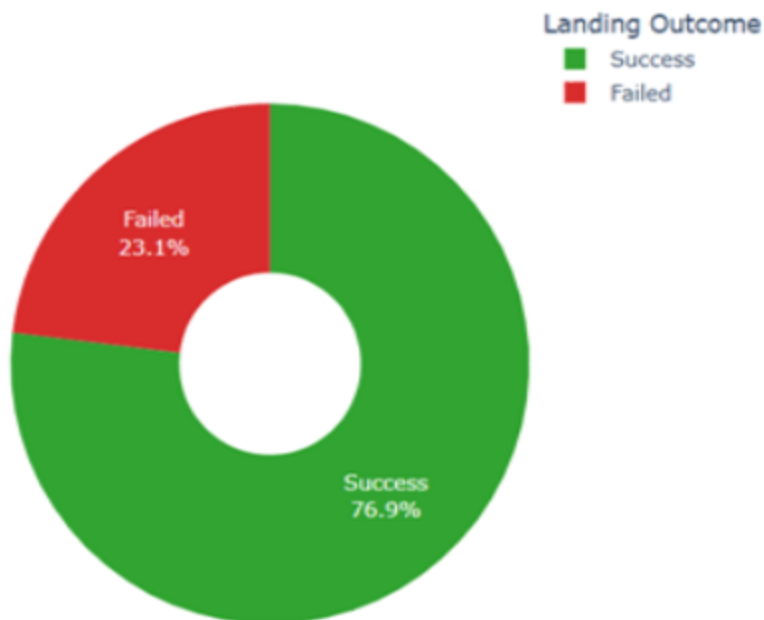
5,000

7,500

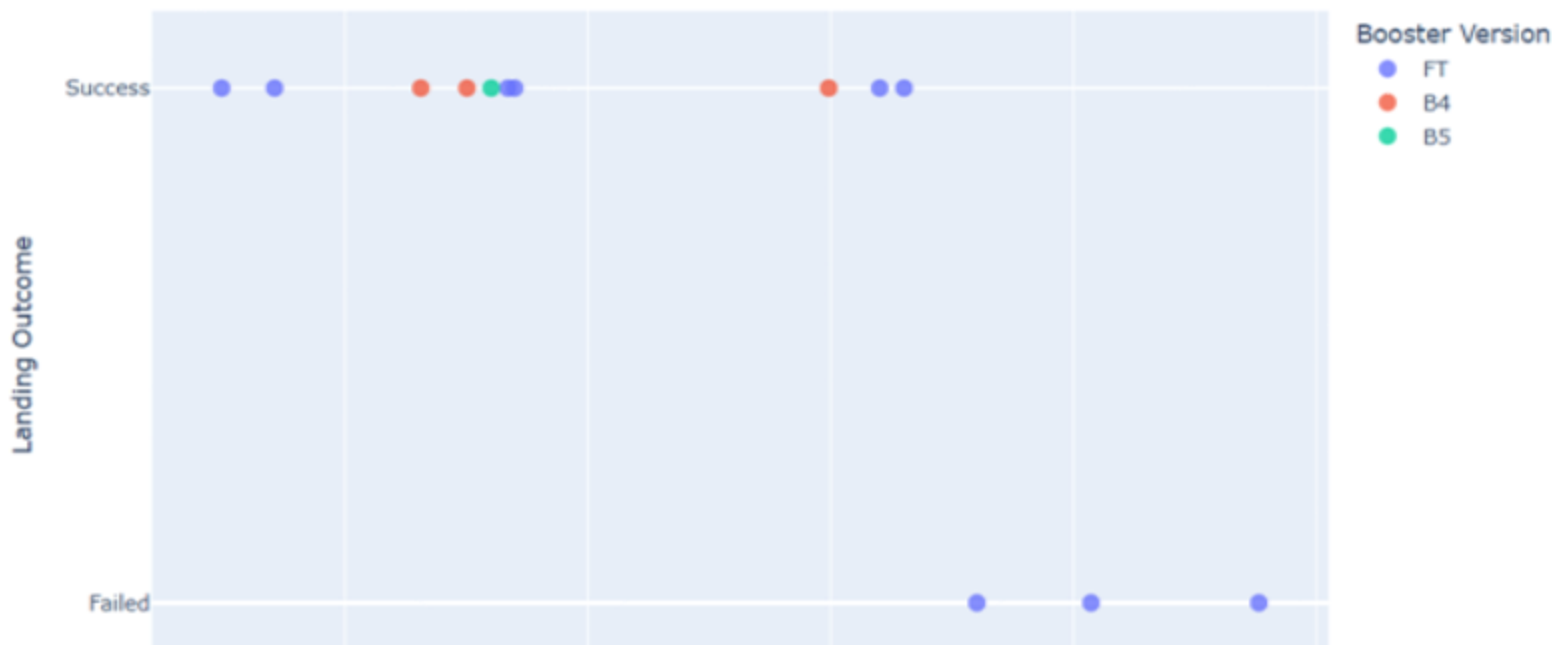
10,000

9600

First-Stage Landing Outcomes — KSC LC-39A



Payload Mass vs. First-Stage Landing Outcome — KSC LC-39A



Example Dashboard

Insight: Landing Outcomes by Site

- Filtering the dashboard to KSC LC-39A shows a substantially higher observed first-stage landing-success proportion than the overall dataset. This demonstrates how the dashboard supports site-specific exploration rather than establishing that launch site alone causes the difference.

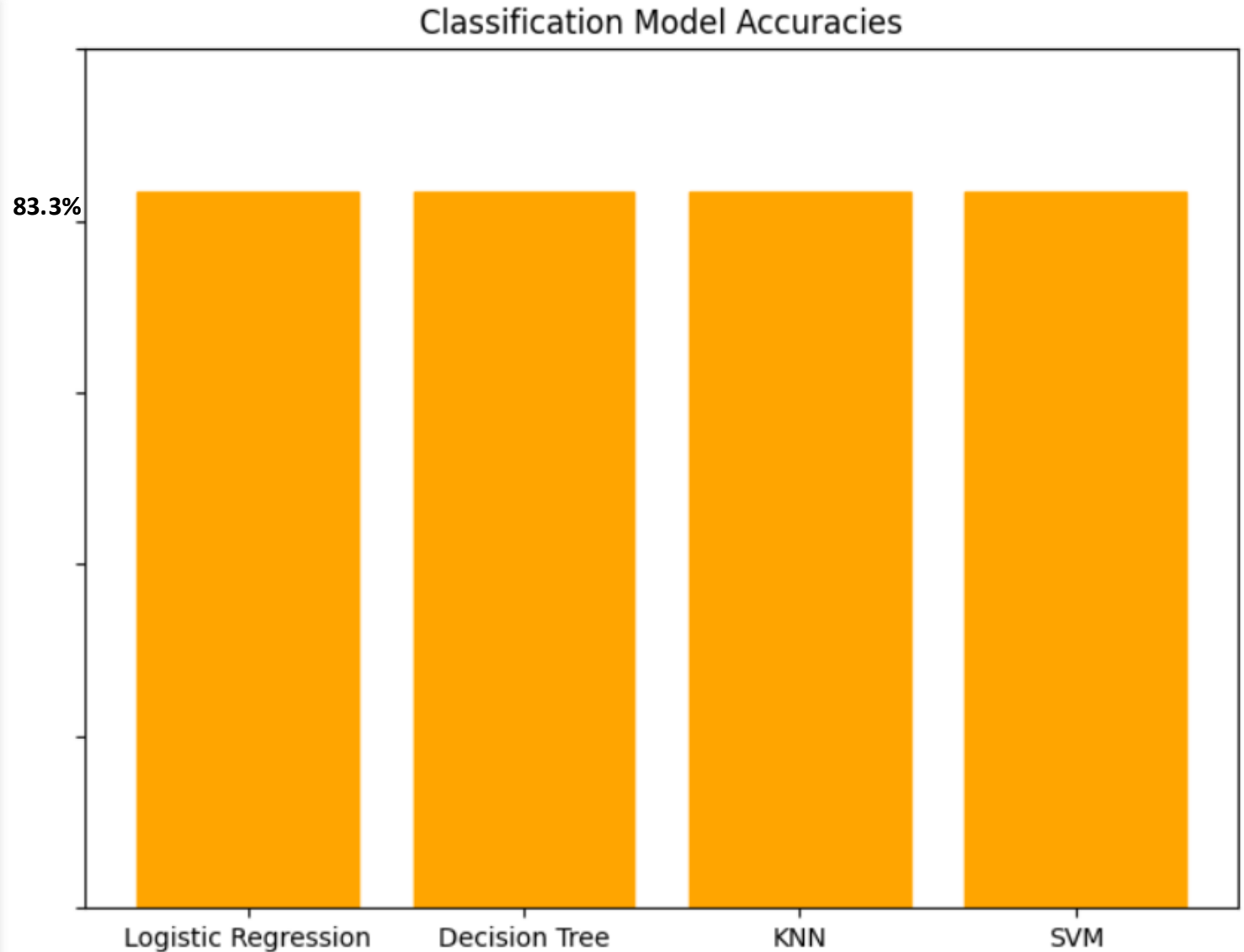
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Key Elements & Findings:

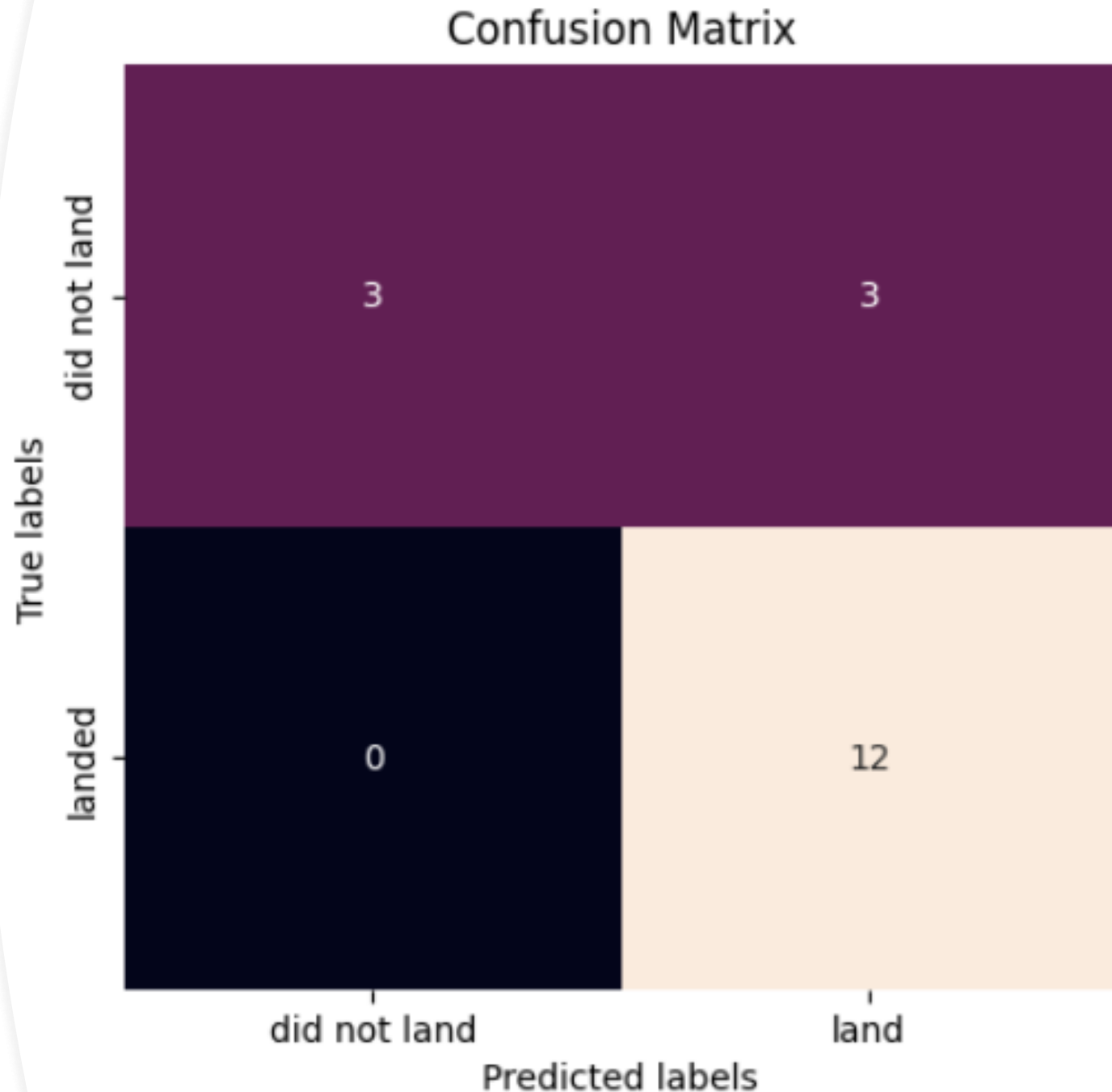
- All four tuned classifiers produced identical predictions on the 18-observation test set, achieving 83.3% accuracy. Performance was asymmetric: all successful landings were identified, while only half of failed landings were correctly detected. Because the test set is small, these results do not establish that the models are generally equivalent.
- Majority-class baseline: 66.7% accuracy (12/18).



Confusion Matrix

Key Elements & Findings

- **Failed landings:** 3 of 6 correctly identified → **specificity = 0.50**
- **Successful landings:** 12 of 12 correctly identified → **recall / sensitivity = 1.00**
- **Precision for successful landings:** 0.80
- **Overall accuracy:** 83.3%
- The models identified every successful landing in the test set, but correctly identified only half of the failed landings. This produced high sensitivity for successful landings but limited specificity for failures.



Conclusions

- Built an end-to-end data science workflow covering **data acquisition, cleaning, SQL analysis, visualization, geospatial analysis, interactive dashboarding, and machine learning.**
- Exploratory analysis showed meaningful associations between **flight history, orbit type, payload mass, launch site, and first-stage landing outcome.**
- Folium analysis added spatial context by examining launch sites and nearby geographic features.
- Logistic Regression, Decision Tree, KNN, and SVM all achieved **83.3% test accuracy** on the 18-observation test set.
- Model performance was asymmetric: successful landings were identified more reliably than failed landings, highlighting the importance of class-specific evaluation.
- Because the dataset and test set are small, results should be viewed as a **demonstration of the modeling workflow rather than a production-ready predictive system.**

Key Insights & Project Contributions

- Combined **API data, web scraping, SQL, geospatial analysis, dashboarding, and classification** within one project.
- Used Folium not only to map launch sites, but also to analyze their relationship to nearby geographic and infrastructure features.
- Distinguished between simple accuracy and **class-specific performance**, identifying weaker detection of failed landings.
- Built an interactive Dash application that allows users to explore landing outcomes by **launch site and payload range**.
- Applied cautious statistical interpretation by treating observed EDA relationships as **associations rather than causal effects**.
- Identified clear opportunities for improvement through larger datasets, richer mission features, and more robust model validation.

Appendix & Technical Resources

Repository Contents

- Data collection via SpaceX API
- Web scraping workflow
- Data wrangling and feature preparation
- Exploratory data analysis
- SQL analysis
- Folium geospatial analysis
- Plotly Dash dashboard
- Classification and model evaluation
- Supporting datasets and visual outputs

Full source code and notebooks:

github.com/alexandbalica/Applied-Data-Science-Capstone

Thank you!

